The macroeconomic landscape

Jean-Félix Brouillette

Session 1

HEC Montréal | MBA

What is macroeconomics?

Macroeconomics deals with...

- Global economic phenomena: the **big picture**
- **Interconnections** between economic agents (e.g., consumers, firms, gov.)
- **Long-run** trends, **short-run** fluctuations, and economic **policy**

Macroeconomics is closely linked to many other fields:

- Politics and geopolitics
- Psychology and sociology
- Finance
- Environment, technology, etc.

Macroeconomics vs. microeconomics

Microeconomics provides great tools to study how individuals make decisions...

...but not how they **aggregate** to the economy as a whole

...and this aggregation, in turn, matters **a lot** for individual decision-making!

When individual rationality leads to collective madness...

- **The paradox of thrift:** everyone saves more in a recession
- **Helicopter money:** too many dollars chasing too few goods
- **The fallacy of composition:** standing up at a hockey game

Why should *you* care about macroeconomics?

The success of any business decision rests partly on **macroeconomic conditions**

Macroeconomic events, forces, and trends…

1. …disrupt **business strategies**
2. …influence **costs, wages, interest rates**, and **competition**
3. …are sources of **risk**, but also of **opportunity**
4. …shape **markets** on a global scale

…and **macroeconomic jargon** will be at the heart of conversations you will have as future managers

A complex and uncertain world

"Who wrangled the best trade deal from Donald Trump?"

"Kevin Warsh's Fed Job Already Looks Impossible"

"Gold rips past $5,000 on fears of global turmoil"

"Les exportations du Québec en forte baisse"

"Mark Carney's speech got the world's attention—but was it the right message?"

"The Big Money in Today's Economy Is Going to Capital, Not Labor"

A complex and uncertain world

“Who wrangled the best trade deal from Donald Trump?”

“Kevin Warsh’s Fed Job Already Looks Impossible”

“Gold rips past $5,000 on ...”

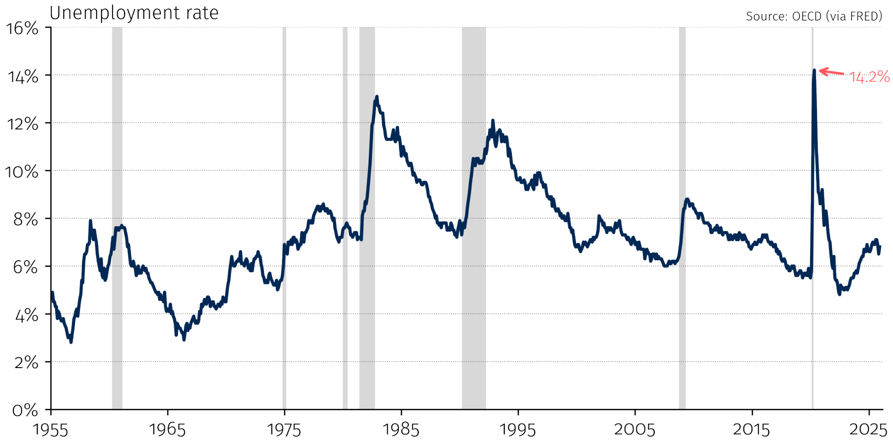
“Les exportations d... Québec ... baisse”

“Mark Carney’s speech got the world’s attention—but was it the right message?”

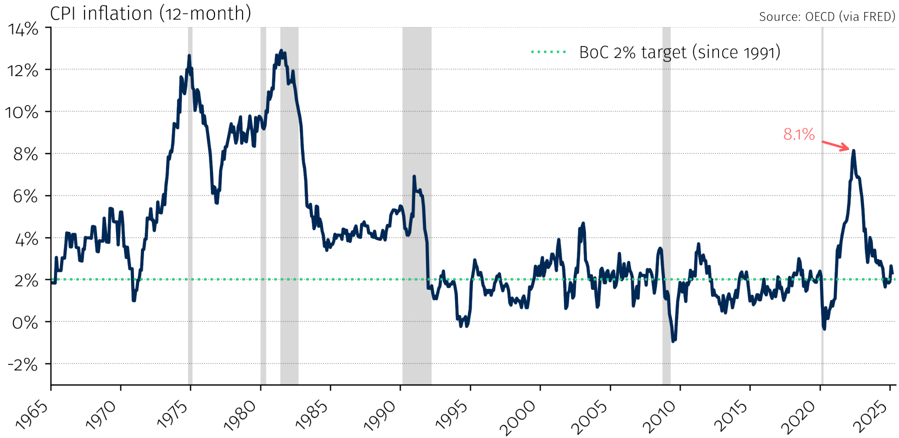
“The Big Money in Today’s Economy Is Going to Capital, Not Labor”

Where are we now?

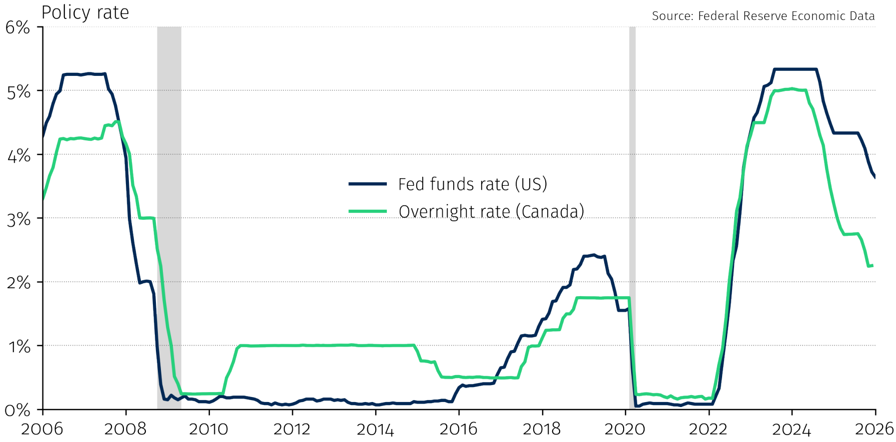
After a very unusual recession…



An old enemy back from the dead...

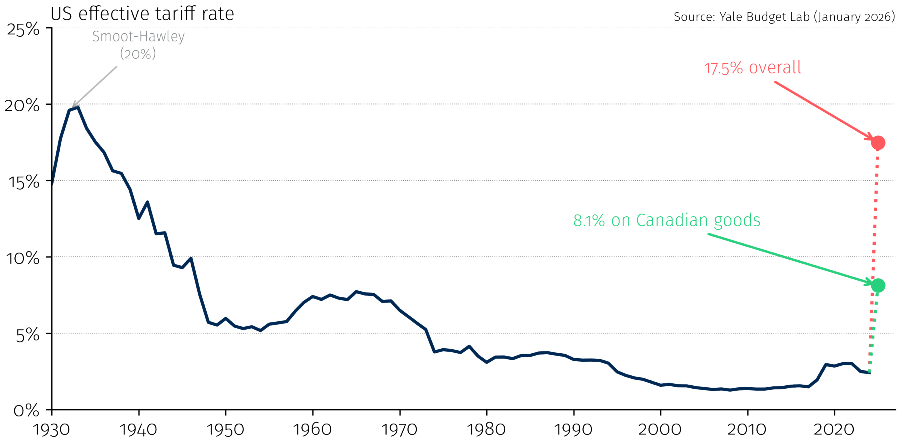


A forceful response…

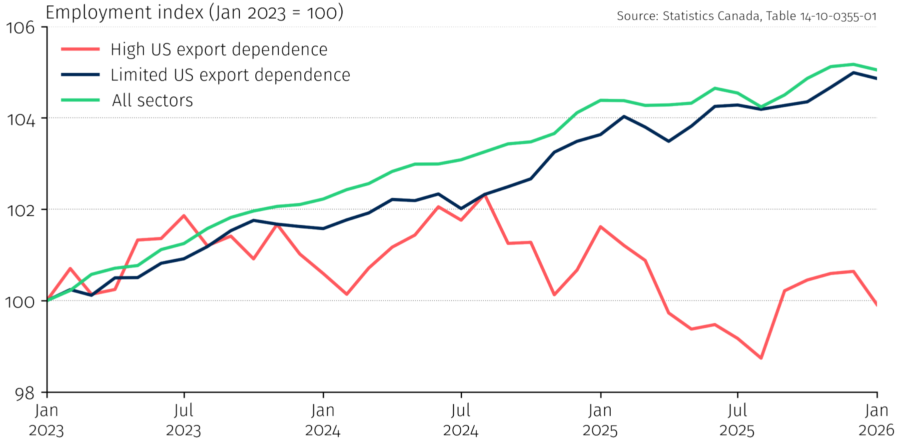


A return to normal…

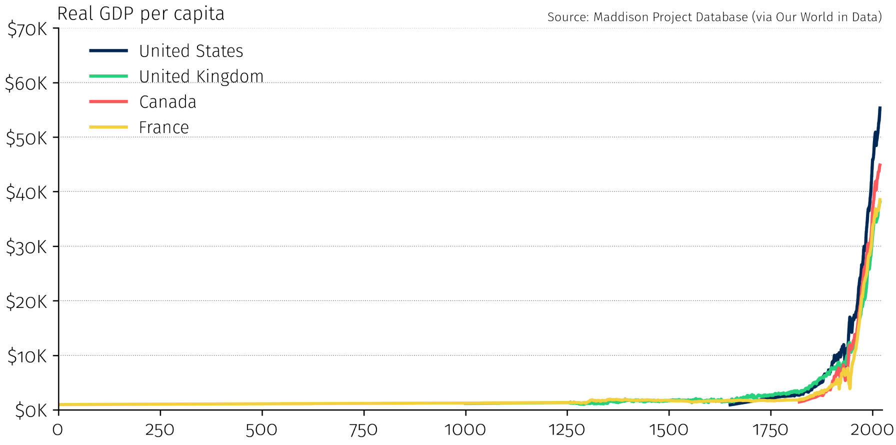
The return of protectionism…



An uncertain future…



What about the distant future?



Outline

1. **Course overview and logistics**
2. Measuring the economy: GDP
3. Measuring inflation
4. Comparing living standards across countries
5. Beyond GDP
6. The macroeconomic landscape in 2026

Course overview

Course objectives

No, I will not make you economists...but by the end of this course, you will:

1. Be **less intimidated** by macroeconomic discussions and jargon
2. Be able to critically interpret **macroeconomic news** (“BS” detector)
3. Be aware of major **macroeconomic trends**
4. Have **objective and coherent** conversations about macroeconomics

What this course is not

Technical. Abstract. Ideological.

Themes to be covered

1. **Long-run economic growth:** why are some countries rich and others poor?
2. **The labor market:** inequality, technology, and AI
3. **Business cycles:** why does the economy boom and bust?
4. **Monetary and fiscal policy:** how governments fight recessions
5. **International trade and imbalances:** (de)globalization, tariffs, etc.

Teaching team

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Pedagogical material

ZoneCours (ZC)

- Slides and articles posted before each class
- Exercises and solutions

Textbook

- Vincent & Yared (Pearson Revel): not mandatory
- ZC: reference for lectures + exercises

Stay on top of macro news

- *The Economist, The Financial Times, The Wall Street Journal, Bloomberg, La Presse Affaires, Reuters, Les Échos, etc.*

Grades

Team project (two parts)	**30%**
Quiz (session 4, ~15 minutes, multiple choice)	**10%**
Final exam (March 30th)	**50%**
Contribution and professionalism	**10%**

Details

Team project details are on ZC. Examples of quiz questions will be provided.

Contribution and professionalism

Criteria:

1. **Attendance and punctuality** — as per program requirements
2. **Active participation** — quality and relevance of interventions
3. **Professionalism** — autonomy, judgment, collegiality

To promote active learning:

1. **Name cards** — please bring yours to every class
2. **No laptops** in class (exception: tablets lying flat)
3. **Limit comings and goings** during class

Outline

Measuring the economy: GDP

What is GDP?

Gross Domestic Product (GDP)

The total **market value** of all **final** goods and services produced **within a country** during a **given period**.

Key words:

- **Market value:** measured in dollars (or another currency)
- **Final:** excludes intermediate goods to avoid double-counting
- **Within a country:** production on Canadian soil
- **Given period:** a flow, not a stock (typically quarterly or annual)

Three ways to measure GDP

They all give the same answer

1. Expenditure

Who is buying?

$$Y = C + I + G + NX$$

- C: consumption
- I: investment
- G: gov. spending
- NX: net exports

2. Income

Who is earning?

- Wages/salaries
- Corporate profits
- Rental income
- Interest income

3. Value added

Who is producing?

- Value added at each stage
- Revenue − int. inputs

Key insight

Every dollar spent is a dollar earned is a dollar of value created

Example: value added avoids double counting

Example

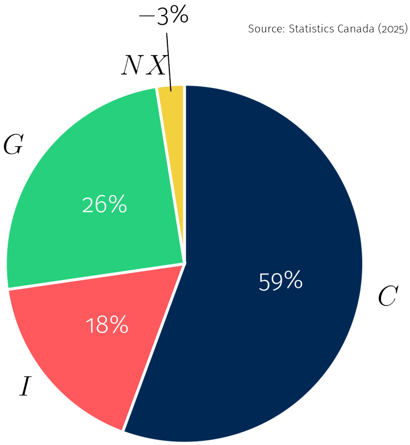
A business hires employees to work in a fertilizer plant and pays them \$0.50 in wages. The business sells the fertilizer to Dwight, a self-employed farmer, for \$0.80. Dwight uses it to produce beets, which he sells to Jim for \$1. Jim eats the beets.

Production		**Income**		**Expenditure**	
Manufacturing	0.80	Wages	0.50	Consumption	1.00
Agriculture	0.20	Profits	0.30		
		Prop. income	0.20		
Total	**1.00**	**Total**	**1.00**	**Total**	**1.00**

Key insight

Production = **value added** (revenue − intermediate inputs), not total revenue. This avoids double counting and ensures all three approaches agree.

The expenditure approach: $Y = C + I + G + NX$



Consumption (C)

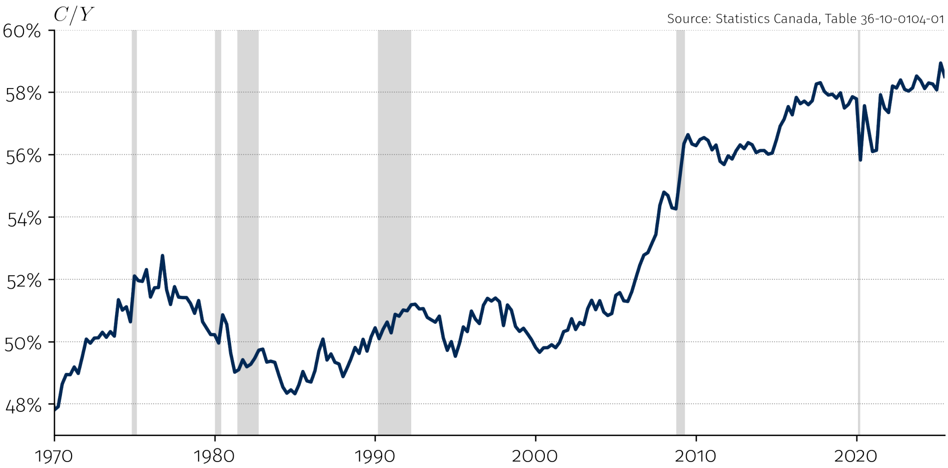
Household spending on final goods and services:

- **Durable goods** — cars, furniture, appliances (long-lasting)
- **Non-durable goods** — food, clothing, gasoline (used up quickly)
- **Services** — rent, healthcare, haircuts, education

Key insight

Services account for over 60% of consumption in Canada. The shift from goods to services is a hallmark of advanced economies.

Consumption share of GDP in Canada



Consumption is the largest component (≈60%), and increasing over time.

Investment (I)

Spending on goods used to produce future output:

- **Business fixed capital** — machines, equipment, factories, software
- **Residential** — new housing construction
- **Inventories** — unsold goods stored in warehouses

Common confusion

In economics, “investment” means **physical capital formation**, not buying stocks or bonds. A firm purchasing a delivery van is investment; buying shares of Tesla is not.

Example: investment

Example

General Motors produces a minivan, which it sells for $1,000 to the Michael Scott Paper Company. The cost of producing it is made up of workers' wages worth $600.

Production		**Income**		**Expenditure**	
Manufacturing	1,000	Wages	600	Investment	1,000
		Profits	400		
Total	**1,000**	**Total**	**1,000**	**Total**	**1,000**

Key insight

A firm buying a capital good = **investment**, not consumption.

Example: inventories (Year 1)

Example

Dunder Mifflin produces 500 tons of white paper worth $400 and stores them in its warehouse while it waits for customers to buy them. The cost of producing them is made up of workers' wages of $500.

Production		Income		Expenditure	
Manufacturing	400	Wages	500	Investment	400
		Profits	(100)		
Total	**400**	**Total**	**400**	**Total**	**400**

Key insight

Unsold goods = **inventory investment**. GDP captures all production, even without a sale.

Example: inventories (Year 2)

Example

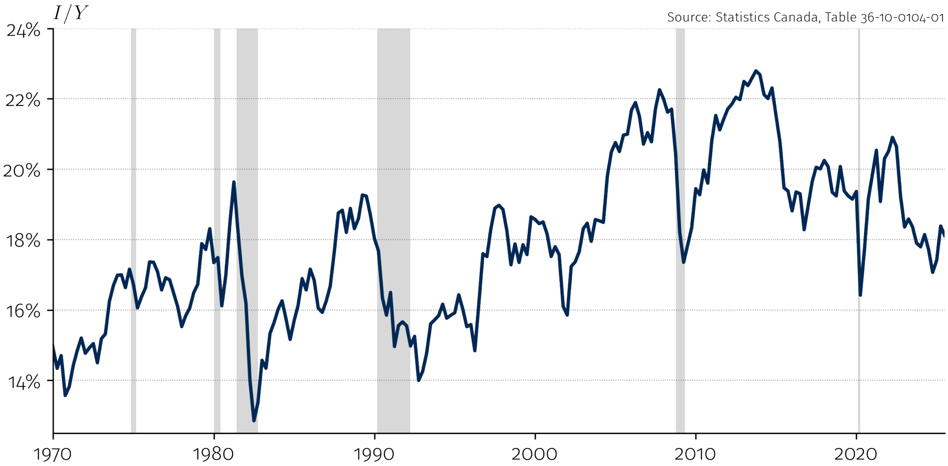
The following year, Dunder Mifflin finally sells its paper worth $400.

Production		**Income**		**Expenditure**	
				Consumption	400
				Investment	(400)
Total	**0**	**Total**	**0**	**Total**	**0**

Key insight

Selling from inventory: C rises, I falls $\Rightarrow$ GDP = 0. Production was counted in year 1.

Investment share of GDP in Canada



Investment is the most **volatile** component — it swings sharply during recessions.

Government spending (G)

Major producer of goods/services: military, education, healthcare, infrastructure.

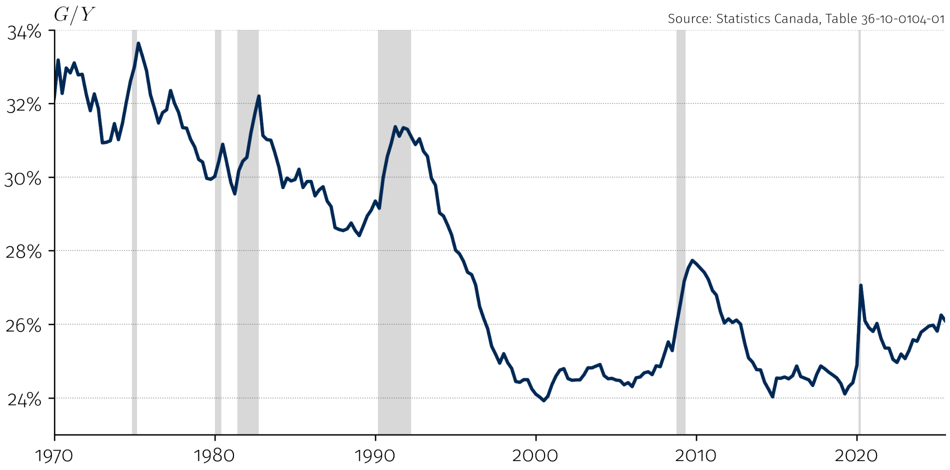
Valued at cost

Often missing market prices, so G measured using **cost of inputs**.

Transfers $\neq$ government spending

Social transfers (UI, pensions, welfare) are **not** in G because they do not involve any new production — they redistribute existing income.

Government spending share of GDP in Canada



Government spending spikes in recessions, but on a long downward trend since the 1990s.

Exports and imports (NX)

C, I, and G include spending on **foreign** goods (imports).

Since GDP measures *domestic* production, we need a correction:

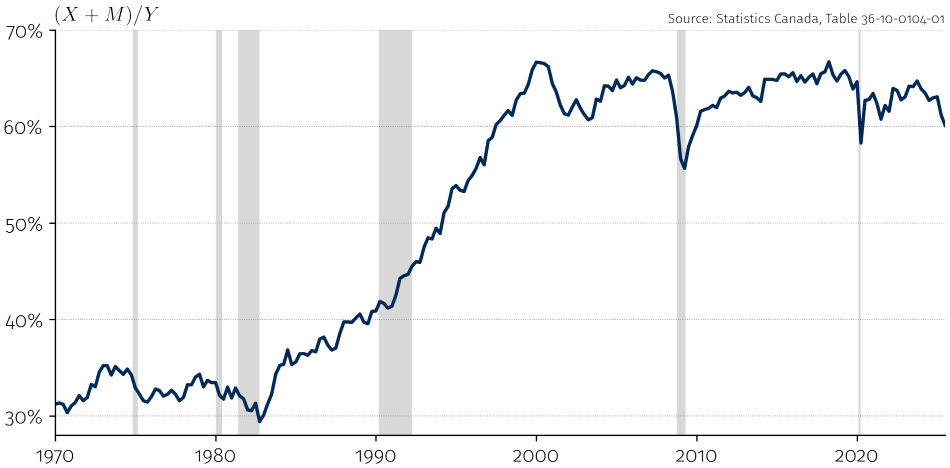
$$Y = \underbrace{C + I + G}_{\text{total spending}} + \underbrace{X - M}_{\text{net exports}}$$

- X (exports): domestic production sold to foreigners
- M (imports): foreign production already counted in C, I, G

Trade openness

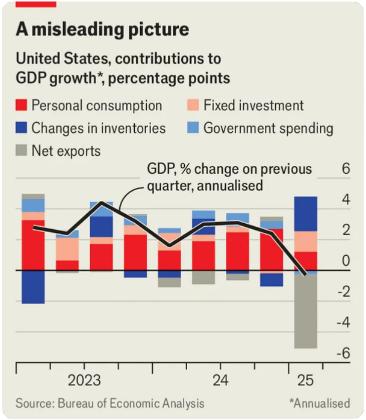
$(X + M)/Y$ measures how important trade is for an economy.

Trade share of GDP in Canada



Canada is a very open economy. Trade peaked with NAFTA (1994) and has slightly declined since.

Training your “BS” detector



Source: *The Economist*, 2025

In Q1 2025, US GDP fell. Many blamed imports: “the trade deficit subtracted 5 pp from GDP growth.”

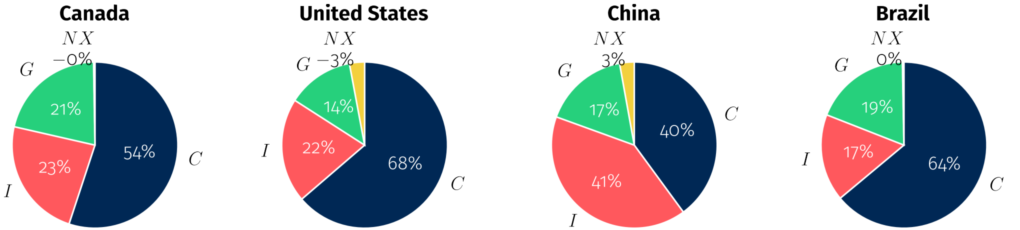
That’s misleading

Imports enter GDP **twice**: once *added* in C, I, or G, and once *subtracted* in NX.

If firms front-load imports before tariffs, inventories surge ($\uparrow I$) while net exports plunge ($\downarrow NX$). The two cancel out.

If consumers hurry to buy dishwashers before tariffs, consumption surges ($\uparrow C$) while net exports plunge ($\downarrow NX$). The two cancel out.

GDP by expenditure: cross-country comparison



Source: World Bank (2024)

Different growth models

China is investment-driven (I = 41%), while the US is consumption-driven (C = 68%). These differences shape trade patterns and growth strategies.

What GDP does *not* measure

GDP only counts **market transactions**. A lot of economic activity is missed:

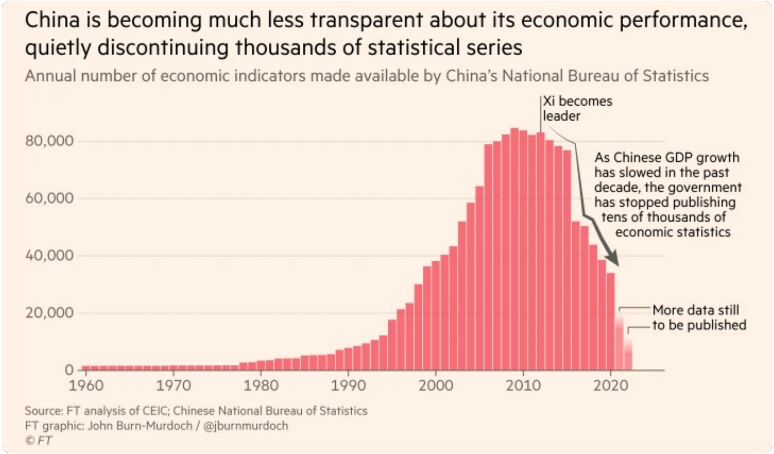
- **Household production** — cooking, childcare, home repairs
- **Volunteer work** — community services, charities
- **Underground economy** — unreported income, informal work

Sex, drugs, and GDP

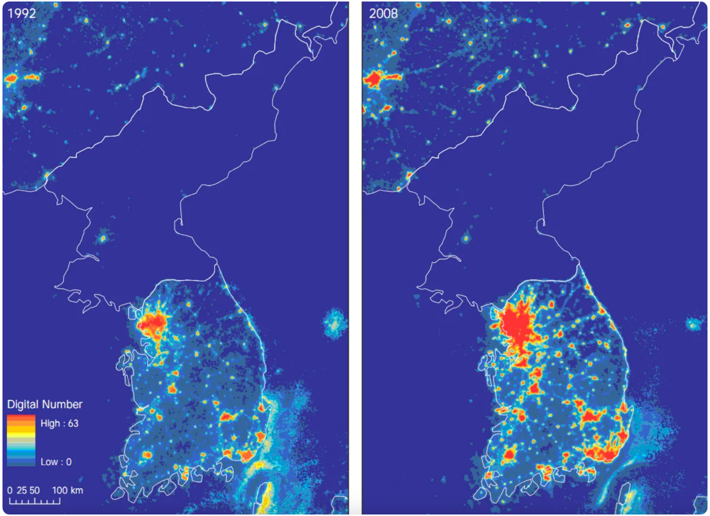
In 2014, EU accounting rules began requiring countries to include estimates of illegal activities (e.g., drugs, prostitution) in GDP. Italy's GDP rose by 18% overnight...

The Economist, 2014

Can we trust GDP figures?



When data is unavailable...



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Measuring inflation

Real vs. nominal GDP

Definition

Nominal GDP = output at *current* prices. **Real GDP** = output at *constant* prices.

Example: nominal GDP grew 5%, prices rose 3% $\Rightarrow$ real GDP grew $\sim$2%.

The **GDP deflator** captures the overall price level:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Key insight

Always use **real** GDP when comparing output over time. Nominal GDP confounds price and quantity changes.

Canadian real GDP since 1960

[Figure: Canadian real GDP, 1960–2025, log scale]
Shade the recessions: 1981, 1990, 2008, 2020

Canada's declining standard of living

GDP growth can mask very different **per capita** dynamics.

- Canada's real GDP per capita has **declined** relative to the U.S. since 2016
- Rapid population growth (immigration) boosted total GDP but **not per capita**
- The U.S.–Canada gap is now at its widest in decades

[Figure: Real GDP per capita]
Canada vs. U.S., 2016–2024]
Source: NBF / StatCan / BEA

Important

Always look at GDP **per capita** when assessing living standards. Total GDP can be misleading.

Measuring inflation

What is inflation?

Inflation

The rate of change in the **overall price level** between two periods.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \%\Delta P$$

where P_t is the price level at time t.

Key question: how do we measure P?

- **GDP deflator:** prices of all goods produced domestically (broad)
- **Consumer Price Index (CPI):** prices of a basket of goods bought by consumers (most commonly reported)

The consumer price index (CPI)

How it works:

- Statistical agencies (Statistics Canada, BLS) survey thousands of prices monthly
- Prices are weighted by a **fixed basket** reflecting typical consumer spending
- Categories: food, housing, transport, clothing, healthcare, recreation, etc.

Headline vs. core inflation:

- **Headline:** all items (volatile)
- **Core:** excludes food & energy (less noisy)

[Figure: Canada CPI inflation]
Headline vs. core, 2005–2026]
with BoC 2% target

Headline vs. core inflation

Headline inflation includes all items — but some are very volatile (food, energy, gasoline).

Core (underlying) inflation strips out extreme price changes to reveal the **trend**.

- Bank of Canada uses CPI-trim and CPI-median
- U.S. Fed focuses on "core PCE" (excludes food & energy)

Key insight

Central banks focus on core inflation for policy decisions — headline inflation is too noisy.

[Figure: CPI and core inflation range, Canada, 2019–2025]
Source: Bank of Canada

Inflation: why it matters for business

Inflation affects:

- **Purchasing power:** wages may not keep up
- **Interest rates:** central banks raise rates to fight inflation
- **Contracts:** long-term deals eroded by unexpected inflation
- **Asset values:** bonds lose value when inflation rises

[Figure: Tariff impact on prices]
Domestic vs. imported, 2024–25]
Source: PriceStats / Cavallo et al.

Business implication

In 2022–23, firms faced a key decision: pass costs to consumers, or absorb them? In 2025, tariffs brought new price pressures.

Comparing living standards across countries

GDP per capita: enormous variation

[Figure: GDP per capita (PPP) since 1950]
US, Canada, Germany, S. Korea, China, India, Nigeria]
Source: Our World in Data / Penn World Table

Comparing GDPs across countries

Why exchange rates can be misleading

Two ways to convert GDP into a common currency:

1. **Market exchange rates**
    - Useful for comparing *economic power* (e.g., trade, geopolitics)
    - Problem: very volatile, and ignores price-level differences
2. **Purchasing Power Parity (PPP)**
    - Adjusts for differences in the cost of living
    - Better measure of *living standards*
    - A dollar buys much more in Lagos than in New York

China vs. United States: it depends how you measure

[GDP at market exchange rate]
US > China

[GDP at PPP]
China > US since ~2016

Which measure to use?

It depends on the question. Geopolitical influence → market rates. Living standards → PPP.

Beyond GDP

What GDP does not capture

GDP is a useful but incomplete measure of well-being. It misses:

- **Inequality:** a country's GDP can grow while most citizens get poorer
- **Non-market production:** household work, childcare, volunteering
- **Environmental costs:** pollution, resource depletion, climate damage
- **Health and well-being:** life expectancy, mental health, leisure time
- **Quality improvements:** your smartphone is infinitely better than a 2005 phone, but GDP struggles to capture this

Growth has lifted millions out of extreme poverty...

[Figure: Share of population in extreme poverty by region, 1981–2020]
Source: Our World in Data / World Bank

...and saved millions of lives

[Figure: Life expectancy at birth, selected countries, 1543–2020]
Source: Our World in Data (Riley, Clio Infra, UN)

But growth alone is not enough

Growth with rising inequality?

- U.S. real GDP per capita has roughly doubled since 1980
- But median household income has barely budged
- The gains have been concentrated at the top

For managers

Inequality shapes consumer demand, labor markets, regulation, and political risk.

[Figure: GDP per capita vs. median income (US)]

The macroeconomic landscape in 2026

Where we stand today

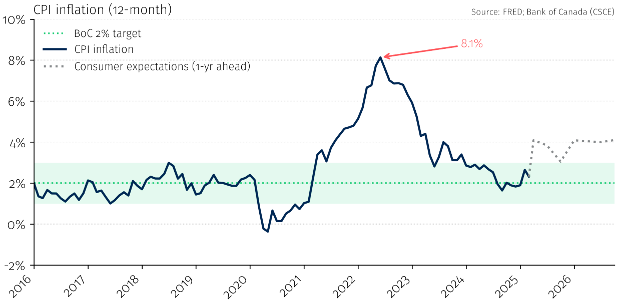
Five themes shaping the global economy

1. **Post-pandemic inflation:** supply shocks → aggressive tightening → disinflation
2. **Interest rate regime shift:** the end of “free money” after near-zero rates
3. **AI and productivity:** will the AI revolution deliver a sustained productivity boom?
4. **Deglobalization:** reshoring, tariffs, and US-China decoupling
5. **Canada:** housing affordability, immigration reversal, and fiscal pressures

Business implication

Each theme connects directly to the analytical tools we will develop in this course.

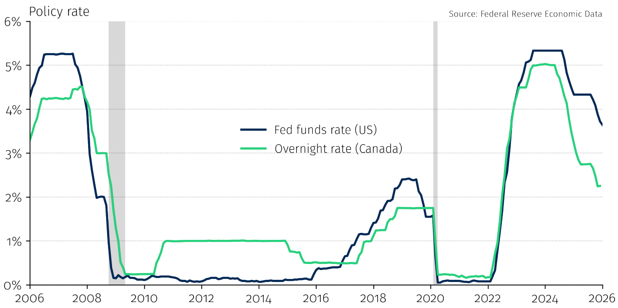
The inflation cycle: a case study in real time



Key insight

In 2021, most economists believed inflation was “transitory.” It wasn’t. Understanding *why* they were wrong is one of the goals of this course.

Interest rates: the end of free money



Business implication

Higher rates mean higher borrowing costs, lower asset valuations, and a different environment for investment and M&A.

AI and the productivity question

The big question:

Will AI deliver a new era of productivity growth, or is it mostly hype?

- **Optimists:** AI is a general-purpose technology (like electricity)
- **Skeptics:** productivity growth has been disappointing for decades despite previous tech booms
- **The data so far:** massive investment, but productivity gains not yet visible in aggregate statistics

We will revisit this in Sessions 2–3.

[Figure: U.S. labor productivity growth]
5-year averages, 1950–2025

Deglobalization and trade tensions

The world is fragmenting:

- US-China decoupling accelerating
- U.S. effective tariff rate at highest in nearly a century (∼17%, Smoot-Hawley levels)
- “Reshoring” and “friend-shoring” as strategy
- Supply chain resilience vs. efficiency

For business

Global supply chains built over decades are being restructured. This creates both risks and opportunities.

[Figure: Progression of Trump tariffs]
US effective tariff rate, 2025
Source: FT / Yale Budget Lab

Canada: the tariff impact in real time

The Canadian economy **contracted** in Q2 2025, driven by a collapse in exports to the U.S.

- Employment in export-dependent sectors **fell sharply** while other sectors grew
- ~75% of Canadian exports go to the U.S.
- Bank of Canada cut rates aggressively in response

Business implication

The GDP expenditure identity in action: a drop in NX directly reduces Y. We will explore these mechanics throughout the course.

[Figure: Canada GDP decomposition]
Q2 2025 contraction]
Source: Bank of Canada

Canada in the global picture

Key challenges for the Canadian economy:

- **Housing affordability:** prices have far outpaced incomes, especially in major cities
- **Immigration policy:** rapid population growth strained housing and services; government reversed course
- **Productivity gap:** Canadian productivity growth has consistently lagged the U.S.
- **Trade exposure:** ~75% of exports go to the U.S. — tariff risks are existential
- **Declining living standards:** real GDP per capita has fallen relative to the U.S.
- **Fiscal pressures:** pandemic deficits pushed public debt to new highs

> **Business implication**
>
> Many of these challenges intersect. We will study each through the lens of macro theory.

Where this course is going

Connecting the themes to the course

Theme	Course sessions
Why are some countries rich?	S2: Long-run growth
AI, inequality, labor shortages	S3: Labor market
Where does inflation come from?	S4: Money & business cycles
Rate hikes, stimulus, debt	S5: Monetary & fiscal policy
Trade, exchange rates, China	S6: Financial markets & international

Key takeaways

1. **GDP** measures the total value of final goods and services produced in a country
2. The expenditure identity ($Y = C + I + G + NX$) is the backbone of macro accounting
3. Always use **real GDP per capita** over time and **PPP-adjusted GDP** across countries
4. **Inflation** (measured by CPI) tracks changes in the cost of living — focus on **core** measures for the trend
5. GDP data is imperfect: think critically about the numbers and who produces them
6. The 2026 landscape: tariff disruptions, declining Canadian living standards, AI, and the end of “free money”

Before next class

Readings:

- Vincent & Yared, Chapters 1–2

To think about:

- Pick a country. Look up its GDP per capita (PPP). What surprised you?
- Find a recent macroeconomic headline. What questions does it raise?